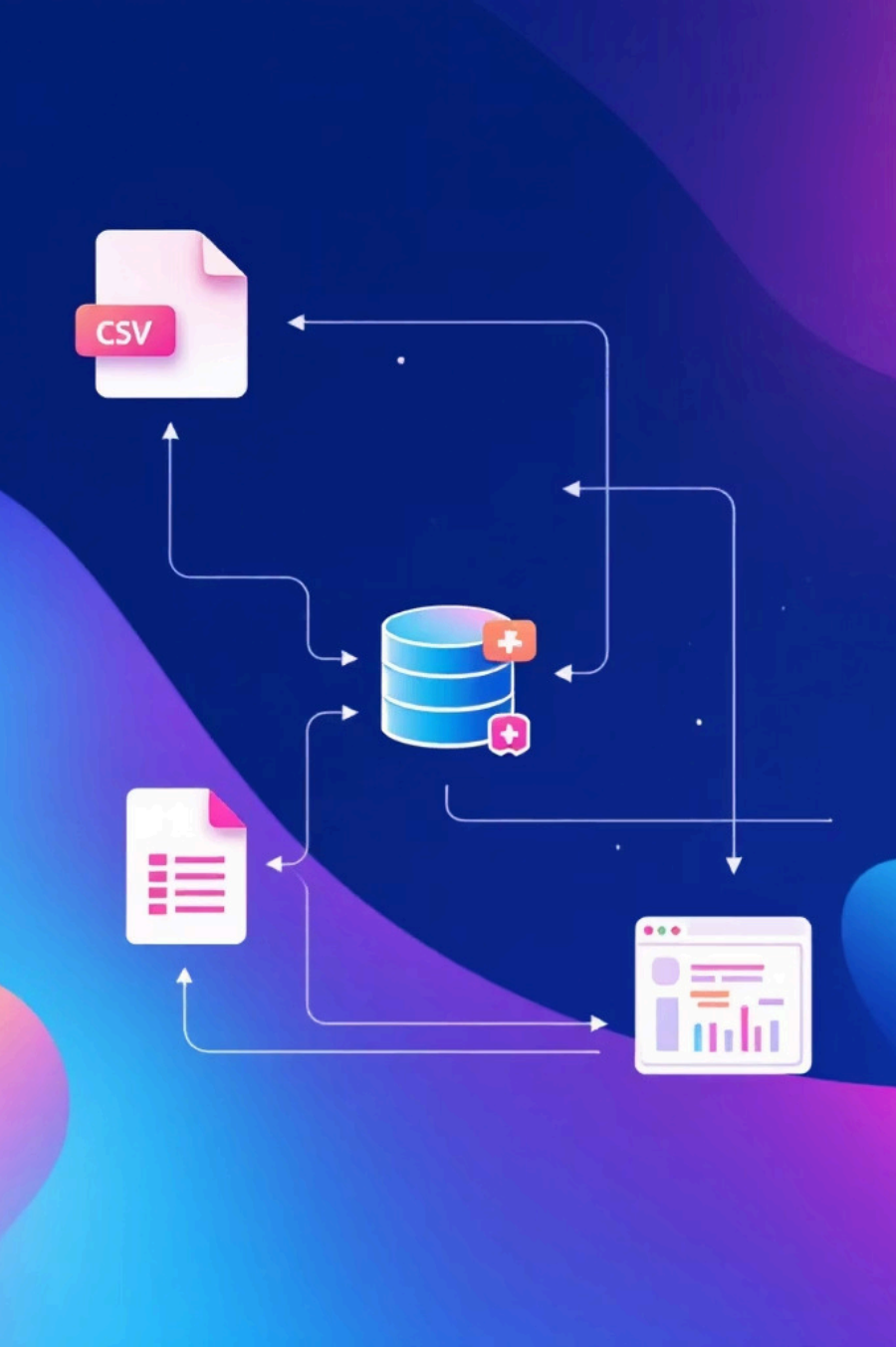




CUSTOMER SHOPPING BEHAVIOR — End-to-End Analytics

Prepared by Fathima Suhaila · 2026

Technologies: Python · PostgreSQL · SQL · Power BI



Executive Summary

Scope

End-to-end study of 3,900 customers: ingestion, cleaning, PostgreSQL storage, 25+ SQL queries, Power BI dashboard.

Pipeline

Python cleaning → SQL ingestion (SQLAlchemy + psycopg2) → PostgreSQL 15 → Power BI interactive reports.

Outcome

Operational dashboard (11 panels) and actionable insights across category, demographics, loyalty, and product quality.

Top-Line KPIs

3,900

Total Customers

Number of unique customers in dataset.

\$59.76

Avg Order Value

Average purchase amount per order.

\$233,0...

Total Revenue

Aggregate revenue across all transactions.

3.75/5

Avg Review Rating

Overall product rating average.

27.5%

Subscription Rate

Approximate share of subscribed customers.

Clothi...

Top Category

Clothing leads with \$104K of revenue.

Project Objectives



Revenue Drivers

Which categories and seasons drive revenue?



Demographic Influence

How do age, gender, and location affect spending?



Subscriptions & Promotions

Impact of subscriptions, discounts, and promo codes on behavior.



High-Value & Loyal Customers

Identify top customers and loyalty patterns.



Payments & Shipping

Preference analysis for checkout optimization.



Product Quality

Which products have low ratings and need fixes?



Seasonal Trends

Revenue trends by season and category.

Data & Cleaning Pipeline



Dataset

3,900 rows × 18 columns (Kaggle). Fields: demographics, product details, amounts, ratings, metadata.

Cleaning Highlights

- BOM fix, snake_case column rename, schema validation
- Type casting, text standardization, boolean mapping
- Median fill for review_rating (37 rows); dedupe → 0 duplicates
- Range clipping (age, amount, rating) and derived columns added

Engineered Features & Ingestion

age_group

Bins: 18–25, 26–35, 36–45, 46–55, 56–65, 66+

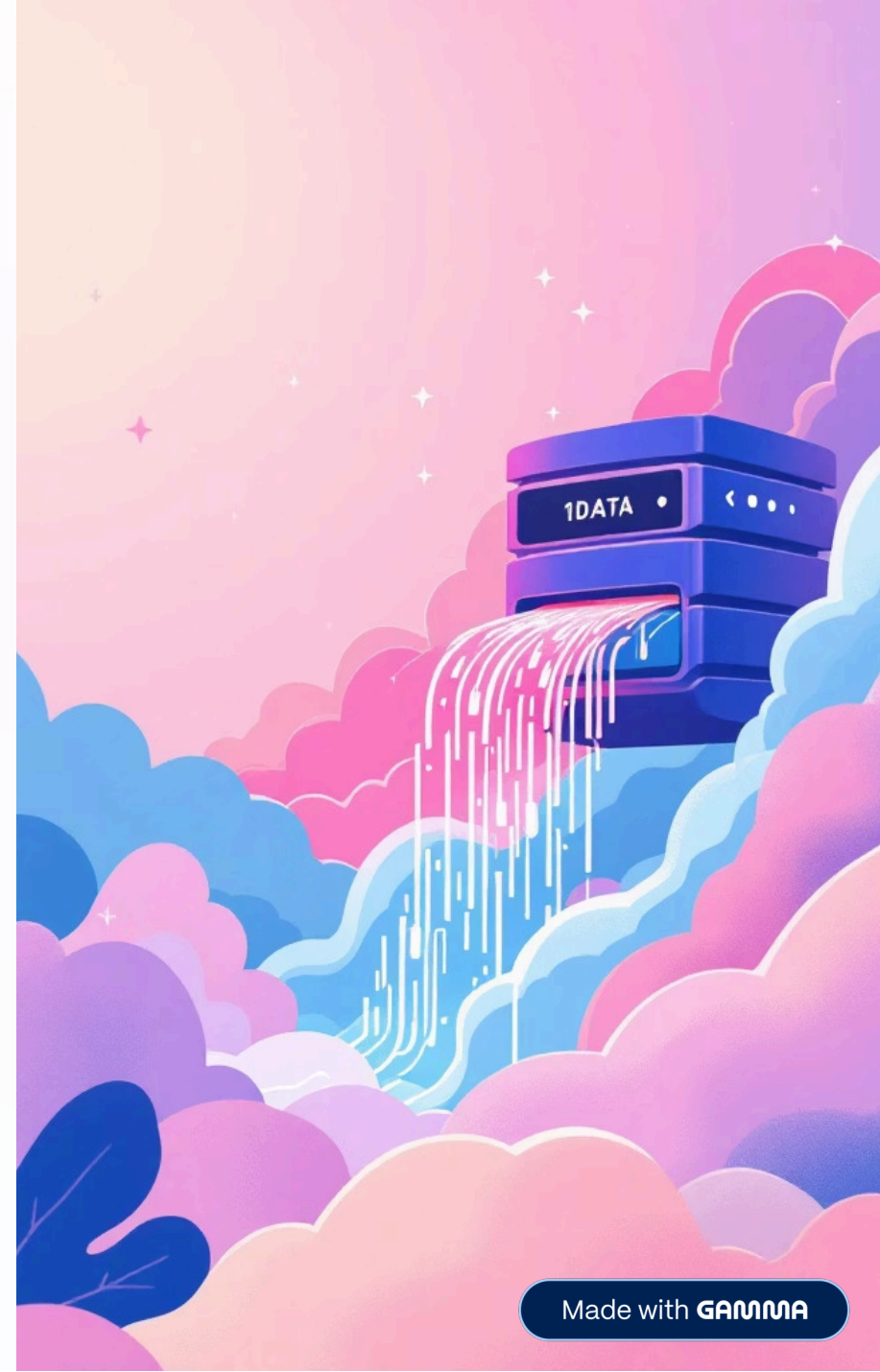
spend_tier

Low <\$40 · Mid \$40–\$70 · High >\$70

is_loyal

Subscribed AND previous_purchases > 25 → Yes/No

Cleaned DataFrame loaded to PostgreSQL 15 via SQLAlchemy (chunksize=500).
Row-count validation and env-var credential support included.

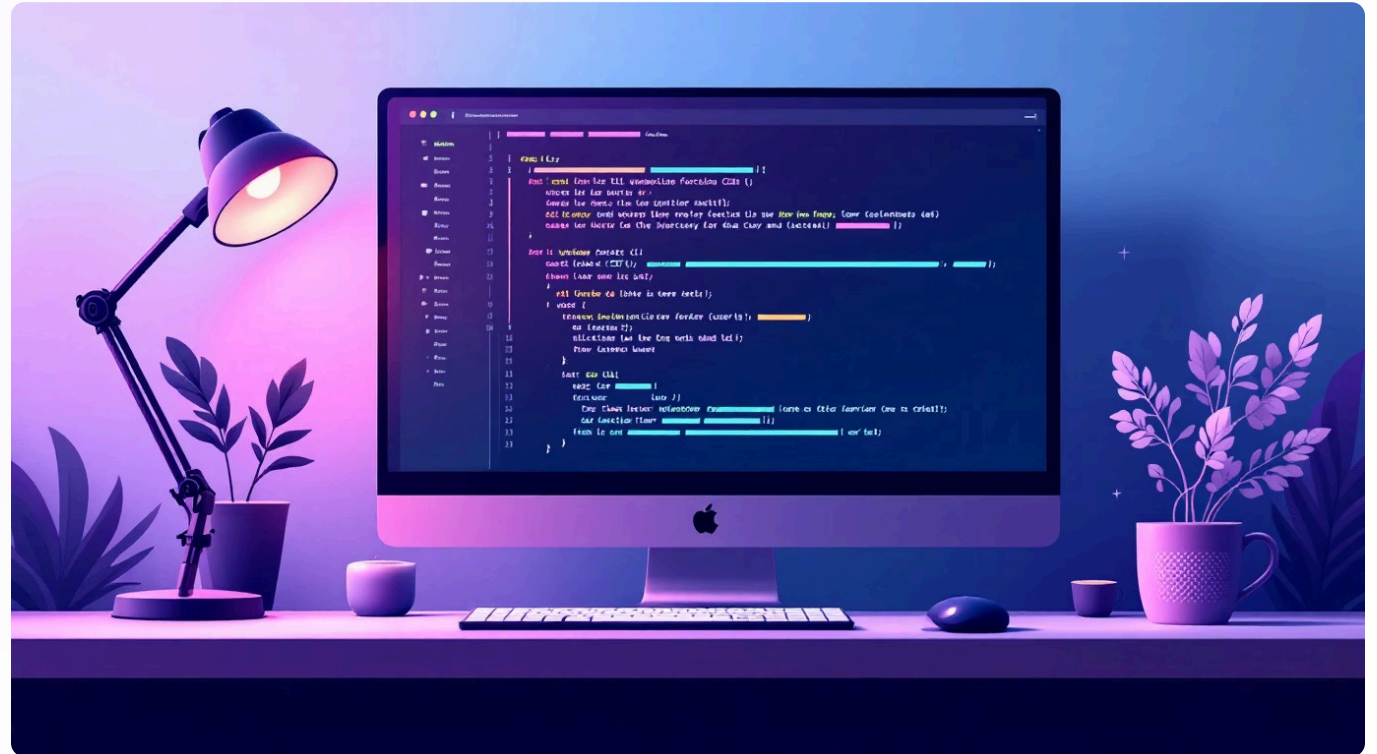


SQL Exploratory Data Analysis

Structure

7 thematic sections · 25+ PostgreSQL queries
using CTEs, window functions, CASE, LAG.

- Overview KPIs (3)
- Demographics (3)
- Product & Category (4)
- Behaviour & Loyalty (5)
- Payment & Shipping (3)
- Review Ratings (3)
- Advanced / Cohort (4)

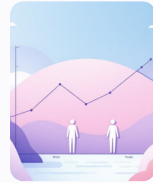


Key Findings



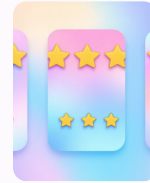
Revenue & Category

Clothing \$104K (44%), Accessories \$74K (32%).
Revenue balanced across seasons (~25% each).
Outerwear underperforms.



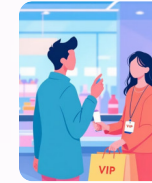
Demographics

Customer base:
68% Male / 32% Female. Core ages 36–55; low engagement under 25. Top 10 states identified for targeting.



Products & Ratings

Top items cluster at 163–171 purchases. Avg rating 3.75/5; low-rated items flagged for quality review.



Loyalty & Subscriptions

Non-subscribers: \$170K vs
Subscribers: \$63K.
Loyal segment shows higher AOV and ratings; promo impact is mixed.

Power BI Dashboard & Visuals



11 Panels

- KPI cards: Total customers, Avg order, Avg rating
- Donut: Revenue by category & by season
- Bar/Column: Age vs Avg Spend, Top Selling Items, Subscription impact
- Heatmap / Matrix: Category × Season
- Shipping & Payment preference panels

Designed for at-a-glance insights with drill-down capability for analysts and stakeholders.

Recommendations & Next Steps

High Priority

Launch female-targeted acquisition campaigns to address 68/32 gender skew.

High Priority

Add subscription perks (exclusive discounts, early access) to increase subscriber AOV.

Medium Priority

Review Outerwear assortment and seasonal marketing to lift the weakest category.

Medium Priority

Address low-rated items identified in EDA to raise average rating toward 4.0.

Low Priority

Target under-25 demographic with brand-building campaigns for long-term growth.

Low Priority

Analyze promo code ROI and restrict blanket discounts to protect AOV.

Project artifacts: code, cleaned CSV, ingestion scripts, SQL queries, Power BI file, README — ready for peer review and executive sharing.